

US Train Tracker

Installation Guide

Mac · Windows · Raspberry Pi · iPhone · Android

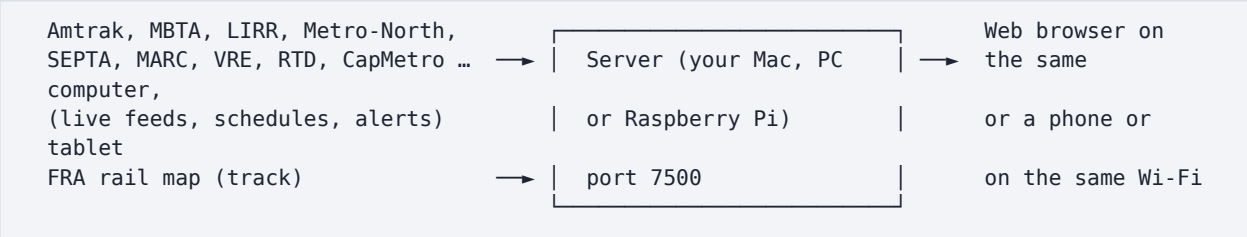
Designed and Created by Project Moon Pi

October 2026 · github.com/ricsne2010/train-tracker

1. How it works

US Train Tracker has two parts:

1. **The server**: a small program that runs on a computer (a Mac, a Windows PC or a Raspberry Pi). It collects live train positions, predictions and alerts from Amtrak and the commuter railroads every 20–60 seconds, along with each railroad's schedule and the federal rail map.
2. **The map**: a web page the server provides. You open it in a web browser on the same computer, or on any phone, tablet or computer on the same Wi-Fi. Trains move on the map as the server receives new positions.



Phones can't run the server themselves: the server is a Python program, and iPhones and Android phones can't run it as an app. A phone shows the map from a server running on a computer at home. For a map that's always available, run the server on a Raspberry Pi that stays on (section 6).

Nothing is sent anywhere except requests to the railroads' public data feeds. There's no account, and the map is only visible on your own network unless you set up remote access (section 7).

2. What you need

	Requirement
Computer for the server	A Mac (macOS 11 Big Sur or newer), a Windows 10 or 11 PC, a Linux computer, or a Raspberry Pi 4 or 5 (a Pi 3B+ works, more slowly)
Python	Version 3.10 or newer, from python.org. 3.12 or newer is recommended: 3.10 stopped getting security fixes in October 2026
Memory	About 150–200 MB while running
Disk space	About 200 MB: the app (~40 MB) and its downloaded maps and schedules (~140 MB)
Internet	Always on while the server runs. The first start downloads about 140 MB; after that it uses about 130 MB an hour (about 3 GB a day if left running all day)
To view the map	A current web browser: Safari, Chrome, Edge or Firefox. Phones and tablets: iPhone or iPad with iOS 15 or newer, or Android with current Chrome
Phones and tablets	Must be on the same Wi-Fi as the server (or connected through remote access, section 7)

The server is light: it sits at almost no CPU between updates. It needs to keep running for the map to stay live; close it and the map stops updating.

3. Install on a Mac

Step 1: Install Python

1. Go to python.org/downloads and click **Download Python 3.x** (the newest version).
2. Open the downloaded `.pkg` file and click through the installer.

You only do this once. If Python is already installed, skip this step: the launcher checks for it.

Step 2: Download US Train Tracker

1. Go to the Releases page.
2. Under **Assets**, click **Source code (zip)**.
3. Open the downloaded zip (Safari usually unzips it for you). You get a folder named like `train-tracker-0.3.0`. Move it somewhere you'll find it, such as your Documents folder.

Step 3: Start it

1. In that folder, double-click **Start Train Tracker.command**.
2. macOS may say it can't verify the developer, because the launcher isn't from the App Store. To allow it: open **System Settings → Privacy & Security**, scroll down to the message about "Start Train Tracker.command", and click **Open Anyway**. (On macOS 14 or older you can instead Control-click the file, choose **Open**, then **Open** again.) You only do this once.
3. A Terminal window opens. The first time, it sets itself up (about a minute), then downloads the rail map and schedules (another minute or two).
4. Your browser opens the map at **http://127.0.0.1:7500** and trains appear.

To stop it, close the Terminal window. **To start it again**, double-click the launcher again: it starts in a few seconds from then on.

Want phones to see it too?

Use **Start Train Tracker (phones too).command** instead. If macOS asks whether Python may accept incoming network connections, click **Allow**. Then see section 5.

4. Install on Windows

Step 1: Install Python

1. Go to python.org/downloads and click **Download Python 3.x**.
2. Run the installer. **On the first screen, tick "Add python.exe to PATH"**, then click **Install Now**.

Step 2: Download US Train Tracker

1. Go to the Releases page.
2. Under **Assets**, click **Source code (zip)**.
3. In your Downloads folder, right-click the zip and choose **Extract All…**. You get a folder named like `train-tracker-0.3.0`.

Step 3: Start it

1. In that folder, double-click **Start Train Tracker.bat**.
2. Windows may show "Windows protected your PC" because the launcher isn't signed. Click **More info**, then **Run anyway**. You only do this once.
3. A black command window opens. The first time, it sets itself up (about a minute), then downloads the rail map and schedules.
4. Your browser opens the map at **http://127.0.0.1:7500**.

To stop it, close the command window.

Want phones to see it too?

Use **Start Train Tracker (phones too).bat**. When Windows Defender Firewall asks about Python, allow it on **Private networks** (your home Wi-Fi). Your Wi-Fi must be set as a private network: Settings → Network & Internet → Wi-Fi → your network → **Private network**. Then see section 5.

5. Use it on an iPhone, iPad or Android phone

1. Start the server with a **(phones too)** launcher (sections 3 and 4), or on a Raspberry Pi (section 6).
2. On the computer, open the map and look at the bottom of the panel for **Open on your phone**. It shows the address to use, such as `http://192.168.1.20:7500`, and a QR code.
3. On the phone, connected to the **same Wi-Fi**, scan the QR code with the camera, or type the address into Safari or Chrome.

The map fits small screens: the panel sits along the bottom; tap ☰ to fold it away.

Put it on your home screen

- **iPhone or iPad (Safari):** tap the **Share** button, then **Add to Home Screen**, then **Add**. The Train Tracker icon opens the map full screen, like an app.
- **Android (Chrome):** tap ⋮, then **Add to Home screen**. The icon opens the map in Chrome. (A full "installed app" window needs a secure https address, such as the remote access in section 7.)

When you're away from your Wi-Fi the icon can't reach the server, unless you've set up section 7.

6. Always on: a Raspberry Pi at home

A Raspberry Pi is a small, quiet, low-power computer that can run the server all the time, so the map is always ready on your phones and computers.

You need: a Raspberry Pi 4 or 5 (2 GB or more), its power supply, a microSD card (16 GB or more), and your home Wi-Fi or a network cable.

Step 1: Set up the Pi

1. On your computer, install **Raspberry Pi Imager** from raspberrypi.com/software.
2. Choose your Pi model and **Raspberry Pi OS (64-bit)** (the Lite version is enough), and your microSD card.
3. When it offers to edit settings, set a **hostname** (`raspberrypi` is fine), a **username and password**, your **Wi-Fi**, and turn on **SSH**. Write the card.
4. Put the card in the Pi and power it on. After a couple of minutes, open Terminal (Mac) or PowerShell (Windows) and connect: `ssh yourusername@raspberrypi.local`

Step 2: Install US Train Tracker

Type these on the Pi, one line at a time. Replace `0.3.0` with the newest version on the Releases page.

```
sudo apt update && sudo apt install -y python3-venv
python3 -m venv ~/train-tracker
~/train-tracker/bin/pip install https://github.com/ricsne2010/train-tracker/releases/
  download/v0.3.0/train_tracker-0.3.0-py3-none-any.whl
~/train-tracker/bin/train-tracker --host 0.0.0.0
```

Wait for the line *On a phone or tablet on this network, open http://…*, then open **http://raspberrypi.local:7500** on your phone or computer. If that works, press **Ctrl+C** on the Pi to stop it, and continue.

Step 3: Start it automatically at power-on

```
curl -LO https://raw.githubusercontent.com/ricsne2010/train-tracker/main/tools/
  train-tracker@.service
sudo cp train-tracker@.service /etc/systemd/system/
sudo systemctl enable --now train-tracker@$USER
```

The map is now at **http://raspberrypi.local:7500** whenever the Pi is on. Check on it with `systemctl status train-tracker@$USER`, and see its messages with `journalctl -u train-tracker@$USER -f`.

To update the Pi later: run the `pip install` line again with the new version, adding `--upgrade`, then `sudo systemctl restart train-tracker@$USER`.

API keys on a Pi: the API keys page only accepts changes from the server's own computer. Either type keys into `~/.train-tracker/.env` on the Pi (see `.env.example` in the project), or add `--allow-remote-settings` to the `ExecStart` line of the service file to manage them from your phone.

7. Seeing it away from home

The simplest safe way is **Tailscale**, a free private network between your own devices:

1. Install Tailscale on the server (Mac, PC or Pi) and on your phone from tailscale.com, and sign in on both with the same account.
2. On the phone, open `http://<the server's Tailscale name>:7500`, for example `http://raspberrypi:7500`.

Please **don't** open port 7500 on your router to the whole internet. The map has no password, and it isn't built to face the open internet.

8. API keys (optional)

Amtrak, MBTA, LIRR, Metro-North, SEPTA, MARC, VRE, RTD Denver and CapMetro work with no setup. Metra, Caltrain, Metrolink and NJ Transit hand out free developer keys. On the map, open the panel and click **API keys…**: each railroad has a **Get a key** link, a **Test** button and a **Save** button. Saved keys start that railroad straight away.

9. Updating and removing

Updating (Mac or Windows): download the new **Source code (zip)** and start it from the new folder. Your API keys and downloaded data are kept in a separate folder (below), so nothing needs moving. Delete the old folder when the new one works.

Removing: delete the train-tracker folder, and the data folder:

- Mac and Raspberry Pi: `~/.train-tracker` (in Finder: Go → Go to Folder → `~/.train-tracker`)
- Windows: `C:\Users\<your name>\.train-tracker`

10. Troubleshooting

What you see	What to do
"Needs Python 3.10 or newer"	Install Python from python.org (step 1). On Windows, reinstall and tick **Add python.exe to PATH**
"Port 7500 is already in use"	Train Tracker is already running (look for its window), or another program uses that port. Close the other window, or start from Terminal with `--port 7600`
The browser didn't open	Open **http://127.0.0.1:7500** yourself
The phone can't open the address	Check the phone is on the same Wi-Fi (not mobile data or a guest network); that you used a **(phones too)** launcher; and that you allowed Python through the firewall. Some routers keep devices apart ("client isolation" or "AP isolation"): turn that off for your home network
`raspberrypi.local` doesn't work	Use the Pi's numbered address instead: run `hostname -I` on the Pi
Map shows but no track	The first start is still downloading the rail map: wait a minute and reload
A railroad shows a red or amber dot	Its feed is down or its key was refused: the panel says why. Feeds come back by themselves
VRE shows 0 trains	Normal outside weekday rush hours
Metra, Caltrain, Metrolink or NJ Transit say "needs an API key"	See section 8

Still stuck? Open an issue on the GitHub page.

US Train Tracker is free software (MIT licence). Train data comes from each railroad's public feeds under their terms; see the Credits in the README. Built with Claude Code (Anthropic). **Designed and Created by Project Moon Pi.**